Active Learning Guide

Simulator: Normal Probability Plot with Mixture of Populations.

With this application, developed using R and Shiny, you have the opportunity to explore how the normal probability plot behaves when the data do not come from a single normal population but from a mixture of several populations with normal distributions. In this case, mixing populations can produce curvatures or clustering of data that are clearly visible in the plot. This application allows you to observe these effects dynamically and in a controlled manner.

The operation of the application is simple. You choose the parameters of the first sample—its mean, standard deviation, and sample size. You can then add a second and third sample, also normal, whose parameters are chosen proportionally to those of the first sample, thus maintaining a defined relationship with the initial values, which makes it easier to understand what is happening when you change each parameter.

Once the three possible samples are established, the application generates the composite sample resulting from merging them, while preserving their identity using color. With this full sample, the normal probability plot is constructed, and you can observe in real time how its appearance changes when modifying the parameters. In this way, you can learn to interpret patterns such as the existence of subgroups and their number, or the presence of anomalous values.